# Leo Zhou

346-859-1973 | zhouleo2007@gmail.com | leozhou8 | leozh0u | Portfolio

## Education

**Rice University** | B.S. in Computer Science | **GPA**: 3.92/4.00 **Expected Graduation May 2028**

- **Relevant Coursework**: Data Structures & Algorithms, Systems Programming, Discrete Math, Linear Algebra

## Experience

**Software Engineer Intern | CanSemi** **June 2026 – August 2026**

- Built an event-driven Python parser pulling measurements from 500+ binary test equipment (ATE) logs
- Designed a normalized PostgreSQL schema and loader ingesting 40 wafer lots (∼12,000 records) in under 2 min
- Wrote a FastAPI service with 8 endpoints over that data, cutting report turnaround ∼75% for 6 engineers
- Built an interactive React and Canvas wafer map with hover and drill-down, retiring the team's Excel heatmaps

**Software Engineer | Inkstone Technologies** **April 2026 – July 2026**

- Built a native iOS reading app in Swift and SwiftUI that adapts every screen to the user's known characters
- Wired Apple Vision OCR into a live camera pipeline overlaying pronunciation on unfamiliar characters only
- Parsed and frequency-ranked a 113,000-entry Chinese dictionary in Python into an indexed SQLite database
- Shipped handwriting input on ML Kit Digital Ink, StoreKit subscriptions, and a WidgetKit home screen widget

**Embedded Software Engineer | Rice Wind Energy** **October 2025 – April 2026**

- Programmed ESP32 firmware in C/C++ reading turbine speed, power output, and blade pitch off live sensors
- Built a six-state safety machine with fault latching and recovery, plus perturb-and-observe power tracking

## Projects

**BlunderNet — Live Chess and Customizable Puzzle Platform** blundernet.com **July 2026**

*Go, AWS, Terraform, Docker, Redis, PostgreSQL, React, PyTorch, C++*

- Built and ran a live chess site for ∼120 users, filtering 3.25M puzzles by rating, theme, phase and length
- Cut uniform random puzzle selection from 1.4 s to 0.9 ms with a precomputed grid and a stored shuffle key
- Kept the Go API stateless with live game state in Redis, moving engine inference onto SQS-fed ONNX workers
- Streamed live moves over WebSockets with Redis pub/sub, so any instance serves any game and none holds state
- Trained the engine in PyTorch from scratch, rewriting its Monte Carlo tree search in C++ for a 1.8× speedup
- Defined the AWS stack in Terraform and load-tested with k6, watching workers autoscale 1 to 4 under backlog
- Traced search latency to cold disk reads at 92% disk utilization, cutting pages touched 55,821 to 20,602

**Vestigo — Evidence-Grounded Geolocation Agent** **Ongoing**

*Python, LLM agent design, tool calling, evaluation harness, pytest*

- Building a photo geolocation agent: a five-step loop of observe, guess, call tools, claim, then resolve
- Measured a no-tool baseline before building anything: 2.6 km median on landmarks, 94.2 km on rural roads
- Replaced distance error with calibration scoring: 88–90% of claims correct at the level claimed, 11% over
- Added a solar-position filter ruling out 49% of the earth, cutting spread between reruns 14,964 km to 537 km
- Gated every answer behind an evidence board that rejects and records any claim citing evidence not on it
- Put one interface over every LLM provider behind a hard budget, routing cheap reads and costly reasoning apart

## Technical Skills

**Languages**: Python, Go, C/C++, Swift, TypeScript/JavaScript, SQL, HTML/CSS
**Backend & Web**: FastAPI, Flask, Node.js, React, SwiftUI, REST APIs, WebSockets, PostgreSQL, Redis, SQLite
**Cloud & Infrastructure**: AWS (ECS, SQS, ALB, RDS), Terraform, Docker, GitHub Actions, Git, k6, pytest
**ML & AI**: PyTorch, ONNX, pybind11, LLM agents, evaluation harnesses, Apple Vision, ML Kit
**Embedded**: ESP32, STM32 (bare-metal), C/C++ firmware, state machines, UART/SPI/I$^2$C

## Leadership & Athletics

- **New Zealand Junior National Fencing Team** — Captain, Coach, Referee; 30+ titles, Junior Worlds
- **Rice Fencing Club** — Manager · **Physics Pathfinders** — Charity President; 600+ students, 22 schools